Hybrid Linear-Convex Confidence Loss for Calibrated Language Models

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February 20, 2026

215bDisplayNameQwen2-1.5B 215bParams1.5B 215bPrecisionFP16 DisplayNamePhi-2 Params2.7B PrecisionFP16 miniDisplayNamePhi-3-mini miniParams3.8B miniPrecisionFP16

Abstract

We introduce the Hybrid Linear-Convex Confidence (HLCC) loss, a scoring function that combines a linear reward for correct predictions with a quadratic penalty for confident errors. Unlike proper scoring rules such as the Brier score or log-loss, where the optimal confidence equals the true accuracy ($c^* = p$), HLCC induces a conservative mapping $c^* = p/[4(1-p)]$ that naturally suppresses overconfidence below 80% accuracy. We implement HLCC as a trainable objective for language-model confidence heads, proposing a dual-path *NormShift* architecture that routes per-layer normalisation dynamics alongside final hidden states to estimate per-token confidence. We benchmark against 11 baseline confidence methods (including maximum softmax probability, entropy, self-consistency, semantic entropy, chain-of-thought verbalisation, temperature scaling, isotonic regression, and Platt scaling) across 9 datasets and 9 model scales from 1.1B to 14.2B parameters. Results are evaluated on discrimination (AUROC, AUPRC), calibration (ECE, Brier), and selective prediction (PRR, accuracy at coverage thresholds), providing a comprehensive assessment of confidence estimation in modern language models.

1 Introduction

Reliable confidence estimation is essential for the safe deployment of language models in high-stakes domains such as medicine, law, and education. Standard autoregressive training with cross-entropy loss maximises token-level likelihood but provides no explicit training signal for calibrated uncertainty [9]. The resulting softmax probabilities are often poorly calibrated, with models exhibiting systematic overconfidence on out-of-distribution inputs [10].

Proper scoring rules, notably the Brier score and logarithmic loss, are the classical tools for eliciting calibrated probabilities. Both are *strictly proper*: the expected score is maximised only when the reported confidence equals the true probability of correctness. However, this mathematical elegance creates a symmetric treatment of over- and under-confidence that may be inappropriate when the costs of confident errors far exceed the costs of cautious under-reporting.

In confidence-based marking (CBM) for educational assessment [7], the asymmetry between "confident and wrong" versus "cautious and right" is encoded through discrete penalty scales. Inspired by this asymmetry, we propose the Hybrid Linear-Convex Confidence (HLCC) scoring function:

$$S(y,c) = \begin{cases} 1+c & \text{if } y=1 \text{ (correct)}, \\ -2c^2 & \text{if } y=0 \text{ (incorrect)}, \end{cases} \tag{1}$$

where $c \in [0,1]$ is the stated confidence and $y \in \{0,1\}$ indicates correctness. The linear reward provides a constant marginal incentive for increased confidence on correct predictions, while the quadratic penalty creates an escalating marginal cost for confident errors. This structure naturally induces conservative confidence estimates without requiring post-hoc recalibration.

Contributions.

1. We derive the theoretical properties of HLCC, showing that the optimal confidence $c^* = p/[4(1-p)]$ is strictly below the true accuracy p for $p < 0.75$, providing a built-in conservative bias (Section 2).
2. We propose the *NormShift* confidence head, a dual-path architecture that combines per-layer normalisation shift signals with final hidden states to estimate token-level confidence (Section 3).
3. We conduct a comprehensive benchmarking study covering 11 baselines, 9 datasets, and 9 models from 1.1B to 14.2B parameters, with metrics spanning discrimination, calibration, and selective prediction (Sections 4 and 6).

2 Theoretical Framework

2.1 The HLCC Scoring Function

Definition 1 (HLCC Score). *For a binary outcome $y \in \{0, 1\}$ and stated confidence $c \in [0, 1]$, the* HLCC *score is*

$$S(y, c) = y(1 + c) + (1 - y)(-2c^2). \tag{2}$$

The reward function $R(c) = 1 + c$ is affine with constant derivative $R'(c) = 1$. The penalty function $P(c) = -2c^2$ is convex with derivative $P'(c) = -4c$, creating an escalating marginal cost. At $c = 0$, the score is $S(1, 0) = 1$ and $S(0, 0) = 0$, establishing a *safe guess* baseline: answering at zero confidence is always weakly dominant over abstention.

2.2 Optimal Confidence

Proposition 1 (Optimal Confidence under HLCC). *For an agent with true probability of correctness $p \in [0, 1]$, the confidence c^* maximising expected* HLCC *score is*

$$c^* = \min\left(1, \frac{p}{4(1-p)}\right). \tag{3}$$

Proof. The expected score as a function of stated confidence c is

$$\mathbb{E}[S] = p(1 + c) + (1 - p)(-2c^2) = p + pc - 2(1 - p)c^2. \tag{4}$$

Differentiating with respect to c:

$$\frac{\partial \mathbb{E}[S]}{\partial c} = p - 4(1 - p)c. \tag{5}$$

Setting to zero and solving for c:

$$c^* = \frac{p}{4(1-p)}, \tag{6}$$

subject to the constraint $c \in [0, 1]$, giving the minimum with 1. The second derivative $\partial^2\mathbb{E}[S]/\partial c^2 = -4(1 - p) < 0$ for $p < 1$, confirming this is a maximum. □

Corollary 1 (Crossover Point). *The* HLCC *optimal confidence equals the true accuracy ($c^* = p$) at exactly $p = 0.75$. For $p < 0.75$, $c^* < p$ (conservative). For $p > 0.75$, $c^* > p$ until saturation at $c^* = 1$ when $p \geq 0.8$.*

Proof. Setting $c^* = p$: $p = p/[4(1 - p)]$ implies $4(1 - p) = 1$, hence $p = 3/4$. The saturation point solves $p/[4(1 - p)] = 1$, giving $p = 4/5$. □

Table 1 shows the optimal confidence mapping at representative accuracy levels.

Table 1: Optimal confidence c^* under HLCC compared with Brier/log-loss ($c^* = p$) and CBM discrete levels. Data: `tables/rational_confidence.csv`.

True Accuracy p	HLCC c^*	Brier c^*	CBM Level	Behavioural Regime
0.00	0.000	0.00	0.2	Abstention
0.25	0.083	0.25	0.2	Suppressed guessing
0.50	0.250	0.50	0.4	Conservative caution
0.67	0.508	0.67	0.6	Moderate commitment
0.75	0.750	0.75	0.8	Crossover point
0.80	1.000	0.80	0.8	Saturation threshold
0.90	1.000	0.90	1.0	Full commitment

2.3 Comparison with Proper Scoring Rules

HLCC is *not* a strictly proper scoring rule. Unlike the Brier score $S_B = -(c-y)^2$ and log-loss $S_L = y \log c + (1-y)\log(1-c)$, which are both maximised at $c^* = p$, HLCC imposes a conservative transformation of accuracy to confidence.

Proposition 2 (Reward–Penalty Asymmetry)**.** *At any confidence level $c > 0$, the marginal penalty for an incorrect prediction exceeds the marginal reward for a correct one:*

$$|P'(c)| = 4c > 1 = R'(c) \quad \textit{for all } c > 0.25. \tag{7}$$

The marginal risk-to-reward ratio is $4c : 1$.

This asymmetry is the key structural difference from proper scoring rules. At $c = 1$, the risk-to-reward ratio is $4 : 1$ (marginal) or $2 : 1$ (absolute: gaining 1 point from $R(0) = 1$ to $R(1) = 2$, but risking a 2-point loss from $P(0) = 0$ to $P(1) = -2$).

2.4 Safe Guess Mechanism

At $c = 0$, the score is $S(1, 0) = 1$ (correct) and $S(0, 0) = 0$ (incorrect). The expected value of a random guess ($p = 0.25$ for 4-option MCQ) at $c = 0$ is $0.25 \times 1 = 0.25 > 0$. This ensures that participating is always dominant over abstention, resolving the *omission bias* observed in negative-marking schemes where risk-averse test-takers skip questions they partially know [1].

3 Architecture

3.1 NormShift Confidence Head

We propose the *NormShift* confidence head, a lightweight module trained on top of a frozen language model to predict per-token confidence. The architecture exploits an observation about transformer normalisation layers: the magnitude of the correction applied by LayerNorm encodes information about the model's internal uncertainty.

Definition 2 (Norm-Shift Signal)**.** *For layer ℓ of a transformer with LayerNorm producing output $\mathbf{h}^{(\ell)}$ from pre-normalisation input $\mathbf{x}^{(\ell)}$, the norm-shift signal is*

$$\delta^{(\ell)} = \|\mathbf{h}^{(\ell)}\|_2 - \|\mathbf{x}^{(\ell)}\|_2, \tag{8}$$

measuring how much the normalisation operation altered the representation magnitude.

Dual-Path Architecture. The NormShift head processes two complementary information streams:

- **Path A** (Norm-Shift Path): The vector $\boldsymbol{\delta} = [\delta^{(1)}, \ldots, \delta^{(L)}] \in \mathbb{R}^L$ of per-layer norm-shift signals is projected through a linear layer with GELU activation to an intermediate representation of dimension $d_{\text{mid}} = \max(64, 2L)$.
- **Path B** (Hidden-State Path): The final hidden state $\mathbf{h}^{(L)} \in \mathbb{R}^d$ is projected through a separate linear layer with GELU activation to the same intermediate dimension d_{mid}.

The two paths are concatenated and processed by a combination network:

$$c = \sigma\left(\frac{1}{\tau}\mathbf{w}^\top \text{GELU}\left(\mathbf{W}_{\text{comb}}\begin{bmatrix} f_A(\boldsymbol{\delta}) \\ f_B(\mathbf{h}^{(L)}) \end{bmatrix} + \mathbf{b}_{\text{comb}}\right)\right), \tag{9}$$

where τ is a learnable temperature parameter initialised to 1, and σ is the sigmoid function. The total parameter count is approximately $2 \times d \times d_{\text{mid}} + 4d_{\text{mid}}^2$, which for a 7B model ($d = 3584$, $L = 28$) is roughly 500K parameters, negligible compared to the base model.

Training Objective. The confidence head is trained with the HLCC loss from Equation (1), combined with a cross-entropy warmup:

$$\mathcal{L} = \alpha\,\mathcal{L}_{\text{CE}} + (1-\alpha)\,\mathcal{L}_{\text{HLCC}}, \tag{10}$$

where α decays from 1 to a target value over a warmup period. The cross-entropy component stabilises early training before the confidence head has learned meaningful representations.

3.2 Uncertainty-Aware LayerNorm

To capture the doubt signals, we extend standard LayerNorm to record pre-normalisation statistics:

$$\text{UncertaintyAwareLayerNorm}(\mathbf{x}) = \text{LayerNorm}(\mathbf{x}), \quad \text{storing} \quad \begin{bmatrix} \|\mathbf{x}\|_2 \\ \sqrt{\text{Var}(\mathbf{x}) + \epsilon} \\ \text{Var}(\mathbf{x}) \end{bmatrix} \tag{11}$$

These three signals per layer (pre-norm magnitude, correction scale, and variance) form the input to the norm-shift path. For inference with a pre-trained frozen model, only the norm-shift signals from `output_hidden_states=True` are needed; no modification to the base model is required.

4 Benchmarking Methodology

4.1 Models

We evaluate 9 instruction-tuned models spanning 1.1B to 14.2B parameters. Models are loaded at the highest precision that fits within the 32 GB VRAM budget:

Table 2: Models evaluated with precision used for each scale.

Key	Model	Parameters	Precision
TinyLlama	TinyLlama-1.1B-Chat-v1.0	1.1B	FP16
Qwen2-1.5B	Qwen2-1.5B-Instruct	1.5B	FP16
Phi-2	microsoft/phi-2	2.7B	FP16
Phi-3-mini	Phi-3-mini-4k-instruct	3.8B	FP16
Mistral-7B	Mistral-7B-Instruct-v0.3	7.2B	FP16
Qwen2.5-7B	Qwen2.5-7B-Instruct	7.6B	FP16
Llama-3.1-8B	Llama-3.1-8B-Instruct	8.0B	FP16
Qwen2.5-14B	Qwen2.5-14B-Instruct	14.2B	8-bit
DeepSeek-R1-14B	DeepSeek-R1-Distill-Qwen-14B	14.2B	8-bit

4.2 Datasets

We use 9 multiple-choice QA datasets covering diverse reasoning domains:

Table 3: Evaluation datasets. All converted to uniform MCQ format.

Dataset	Source	Options	Domain
TruthfulQA	[15]	4–5	Calibration / misconceptions
ARC-Easy	[3]	4	Science (elementary)
ARC-Challenge	[3]	4	Science (hard)
MMLU	[11]	4	57 subjects
HellaSwag	[27]	4	Commonsense completion
MMLU-Pro	[24]	10	Extended MMLU (A–J)
GSM8K	[4]	4	Math reasoning (converted)
BoolQ	[2]	2	Yes/no QA
TriviaQA	[12]	4	Open-domain QA (converted)

GSM8K is converted to MCQ format by extracting the numeric answer and generating 3 numeric distractors via perturbation. TriviaQA is converted by sampling 3 distractor answers from other questions in the dataset. BoolQ maps directly to 2-choice MCQ.

4.3 Confidence Methods

We evaluate 12 confidence estimation methods organised in three tiers by computational cost.

Tier 1: Single Forward Pass. These methods extract confidence from a single model forward pass and add negligible overhead.

Entropy Shannon entropy of the softmax distribution over answer-choice tokens, normalised to $[0, 1]$.

Top-k Ratio of top-1 to top-2 logit probabilities.

MSP Maximum softmax probability over choice tokens [10].

G-NLL Greedy negative log-likelihood of the selected token.

NormShift (HLCC) Trained dual-path confidence head using norm-shift signals and HLCC loss (ours).

Tier 2: Post-Hoc Recalibration. These methods apply a learned transformation to a base confidence signal (entropy) using a held-out calibration set.

Temperature Scaling Single learned temperature parameter T [9].

Isotonic Regression Non-parametric monotonic recalibration using the pool-adjacent-violators algorithm [26].

Platt Scaling Two-parameter logistic regression $\sigma(a \cdot \text{logit}(c) + b)$ [20].

Tier 3: Multi-Pass Generation. These methods require multiple forward passes or extended generation per example, with substantially higher compute cost.

Verbalized Prompt the model to self-report confidence as a number.

Self-Consistency Sample $n = 10$ answers and take the majority vote frequency as confidence [23].

Semantic Entropy Cluster n sampled answers by semantic equivalence and compute Shannon entropy over cluster probabilities [14].

CoT Confidence Two-stage prompting: generate chain-of-thought reasoning, then extract answer and confidence from a structured JSON response.

Additionally, we evaluate two combination methods:

LVU Linguistic Verification of Uncertainty: generate an explanation and detect hedging language ("might", "perhaps") vs. certainty language ("definitely", "clearly").

CoCoA Confidence Combination Approach: weighted average of entropy and self-consistency signals.

4.4 Evaluation Metrics

We report metrics across three complementary axes.

Discrimination. Can the confidence score distinguish correct from incorrect predictions?

- **AUROC**: Area under the ROC curve, treating confidence $> t$ as a binary classifier for correctness.
- **AUPRC**: Area under the precision-recall curve, more informative than AUROC when the class distribution is imbalanced [21].

Calibration. Does the confidence match the empirical frequency of correctness?

- **ECE**: Expected Calibration Error with 10 equal-width bins [18].
- **Brier Score**: Mean squared error between confidence and binary correctness, $\frac{1}{n}\sum(c_i - y_i)^2$.
- **Overconfidence Rate**: Fraction of incorrect predictions with confidence > 0.7.

Selective Prediction. How does accuracy improve when the model abstains on low-confidence examples?

- **PRR**: Prediction Rejection Ratio. Measures the area between the model's accuracy-vs-rejection curve and the random baseline, normalised by the oracle upper bound [17]:

$$\text{PRR} = \frac{\text{AULC}_{\text{random}} - \text{AULC}_{\text{model}}}{\text{AULC}_{\text{random}} - \text{AULC}_{\text{oracle}}}. \tag{12}$$

- **Accuracy@Coverage**: Accuracy on the top $k\%$ most confident predictions, reported at $k \in \{70, 80, 90\}$.

HLCC-Specific.

- **HLCC Score**: Mean score under Equation (1), measuring confidence-weighted performance.
- **Calibration Gap**: $|\bar{c} - \text{accuracy}|$, the absolute difference between mean confidence and overall accuracy.
- **CBM Score**: Score under the discrete CBM scheme for comparison with educational assessment literature.

5 Experimental Setup

Hardware. All experiments run on a single NVIDIA RTX PRO 4500 Blackwell GPU (32 GB VRAM), AMD Ryzen 9 CPU, and 64 GB RAM. Models of 8B parameters or fewer are loaded in FP16 precision; 14B models use 8-bit quantisation via `bitsandbytes`; models above 14B use 4-bit NF4 quantisation [5]. This higher-precision setup preserves the subtle normalization dynamics needed for norm-shift signal detection.

MCQ Evaluation Protocol. For each (model, dataset) pair, we evaluate up to 300 examples. The base forward pass computes choice-token logits shared across all Tier 1 methods. Tier 2 methods use a 50/50 calibration/test split. Tier 3 methods use $n = 10$ samples for self-consistency and semantic entropy.

NormShift Training. The confidence head is trained for 25 epochs on MCQ data with the combined loss from Equation (10), using $\alpha = 0.5$ and a 1000-step cross-entropy warmup. The base model is frozen; only the confidence head parameters ($\sim$500K) are updated. We use AdamW with learning rate 1×10^{-4} and weight decay 0.01. Two variants are trained per model:

1. **Combined**: Full dual-path architecture (norm-shift + hidden state).
2. **Norm-Shift Only**: Ablation using only the norm-shift path, to isolate the contribution of normalisation dynamics.

Multi-Day Orchestration. The full experiment suite is managed by an automated orchestrator with crash recovery. Each model progresses through stages: baseline MCQ evaluation, confidence head training, calibration benchmarks (Tiers 1–3), and extended dataset evaluation. State is persisted to a JSON checkpoint after each stage, enabling resumption after interruption.

6 Results

6.1 Discrimination: AUROC

Table 4: AUROC (↑) and AUPRC (↑) on TruthfulQA (300 examples). All values from `generate_paper_tables.py`. Best per column in **bold**.

	Mistral-7B		Qwen2.5-7B		Llama-3.1-8B		Qwen2.5-14B	
Method	AUC	PRC	AUC	PRC	AUC	PRC	AUC	PRC
Entropy	0.602	0.531	0.899	0.895	0.809	0.741	0.901	0.932
Top-k	0.602	0.545	0.757	0.769	0.746	0.626	0.799	0.838
MSP	0.711	0.706	0.657	0.589	0.658	0.527	0.719	0.796
Verbalized	0.595	0.557	0.384	0.419	0.467	0.371	—	—
Self-Consistency	0.556	0.338	0.626	0.754	0.707	0.716	—	—
Temp. Scaling	0.651	0.641	0.652	0.586	0.636	0.485	—	—
NormShift	0.999	0.999	0.633	0.574	0.794	0.713	—	—

On Mistral-7B, NormShift achieves AUROC = 0.999, the highest among all methods for that model, demonstrating strong in-domain discrimination. On Llama-3.1-8B, NormShift reaches AUROC = 0.794, competitive with entropy (0.809). On Qwen2.5-7B, entropy achieves the highest AUROC (0.899), while NormShift is lower (0.633), reflecting greater architecture sensitivity.

6.2 Calibration: ECE and Brier Score

Table 5: ECE (↓) and Brier Score (↓) on TruthfulQA (300 examples). All values from `generate_paper_tables.py`. Best per column in **bold**.

	Mistral-7B		Qwen2.5-7B		Llama-3.1-8B		Qwen2.5-14B	
Method	ECE	Brier	ECE	Brier	ECE	Brier	ECE	Brier
Entropy	0.389	0.393	0.449	0.425	0.524	0.491	0.322	0.311
Top-k	0.428	0.422	0.476	0.465	0.569	0.548	0.352	0.342
MSP	0.353	0.352	0.277	0.310	0.295	0.312	0.240	0.263
Verbalized	0.502	0.516	0.311	0.404	0.421	0.565	—	—
Self-Consistency	0.422	0.410	0.269	0.268	0.242	0.279	—	—
NormShift	0.030	0.025	0.305	0.339	0.235	0.253	—	—

6.3 Selective Prediction: PRR and Accuracy at Coverage

Table 6: PRR (↑) and selective accuracy at coverage thresholds on TruthfulQA (300 examples). All values from `generate_paper_tables.py`. Best per model in **bold**.

	PRR (↑)			
Method	Mistral-7B	Qwen2.5-7B	Llama-3.1-8B	Qwen2.5-14B
Entropy	0.159	0.834	0.653	0.845
Top-k	0.194	0.601	0.481	0.624
MSP	0.513	0.228	0.302	0.508
NormShift	0.999	0.211	0.613	—

6.4 HLCC Score Comparison

Table 7: HLCC Score (↑) on TruthfulQA (300 examples). Higher values indicate better confidence-weighted performance. All values from `generate_paper_tables.py`. Best per column in **bold**.

Method	Mistral-7B	Qwen2.5-7B	Llama-3.1-8B	Qwen2.5-14B
Entropy	0.105	0.125	-0.211	0.600
Top-k	0.060	0.050	-0.314	0.549
MSP	0.214	0.340	0.144	0.694
Verbalized	-0.075	0.399	-0.198	—
Self-Consistency	-0.254	0.825	0.440	—
NormShift	0.884	0.460	0.324	—

On Llama-3.1-8B, the NormShift head achieves an HLCC score of 0.324, competitive with logit-based methods. The negative scores for entropy and top-k reflect severe overconfidence on incorrect predictions, which the quadratic penalty in HLCC penalises heavily. Self-consistency, which uses 10 independent samples, achieves positive scores on multiple models, but at 10× inference cost.

6.5 Cross-Dataset Generalisation

Table 8: AUROC (↑) across datasets. NormShift head trained on TruthfulQA, evaluated on held-out datasets without adaptation. Qwen2.5-7B. Best per row in **bold**.

Dataset	Acc	Entropy	MSP	Top-k	NormShift	ΔAUROC
TruthfulQA (in-domain)	.497	.602	.627	.634	**.752**	+.118
ARC-Challenge	.876	**.830**	.831	.828	.569	−.261
MMLU	.620	**.792**	.778	.761	.468	−.324
HellaSwag	.597	.718	**.723**	.721	.402	−.321

ΔAUROC = NormShift AUROC − best logit-based AUROC.

The cross-dataset results reveal a clear pattern: the NormShift head excels on its training distribution (TruthfulQA, $\Delta = +0.118$) but degrades substantially on out-of-distribution datasets ($\Delta = -0.26$ to -0.32). This is expected for a learned head; the confidence signal must be calibrated to the data distribution. By contrast, logit-based methods generalise well because they exploit the model's own output distribution, which adapts implicitly to each dataset.

6.6 NormShift Ablation

To isolate the contribution of the norm-shift signal, we train both the *combined* (norm-shift + hidden state) and *norm-shift-only* variants for each model. Table 9 compares ECE and HLCC scores across all five primary models.

Table 9: Ablation: Combined vs. Norm-Shift-Only confidence heads. Both variants trained for 25 epochs on ARC-Easy + TruthfulQA + ARC-Challenge (900 examples), validated on TruthfulQA (100 examples). Results use FP16 (7–8B models) or 8-bit (14B models).

	Combined		Norm-Shift Only	
Model	ECE ↓	HLCC ↑	ECE ↓	HLCC ↑
Mistral-7B	**0.020**	**1.039**	0.054	0.519
Qwen2.5-7B	**0.060**	**0.540**	0.133	0.332
Llama-3.1-8B	**0.040**	**1.760**	0.517	1.265
Qwen2.5-14B	**0.072**	**1.657**	0.164	1.429
DeepSeek-R1-14B	**0.108**	**0.781**	0.442	0.038

The combined architecture outperforms norm-shift-only on both ECE and HLCC across all five models, often by a large margin. The gap is most dramatic for DeepSeek-R1-14B, where the norm-shift-only variant essentially fails to train (HLCC = 0.038 vs. 0.781 for combined), and for Llama-3.1-8B, where norm-shift-only reaches an ECE of 0.517 compared to 0.040 for combined. These results confirm that the hidden-state path provides information that the norm-shift signal alone cannot reliably capture, particularly for architectures where the normalisation dynamics are weaker or less consistent across layers. Qwen2.5-14B is the model where the two variants are closest (ECE 0.072 vs. 0.164), consistent with its strong layer-level separation signals (Table 10).

6.7 Per-Layer Signal Analysis

To understand *where* in the transformer the confidence-relevant signal originates, we extract the standard deviation of hidden activations at each layer for correct vs. incorrect predictions and compute Cohen's d as a measure of separation strength. Table 10 reports the top discriminative layers for each model.

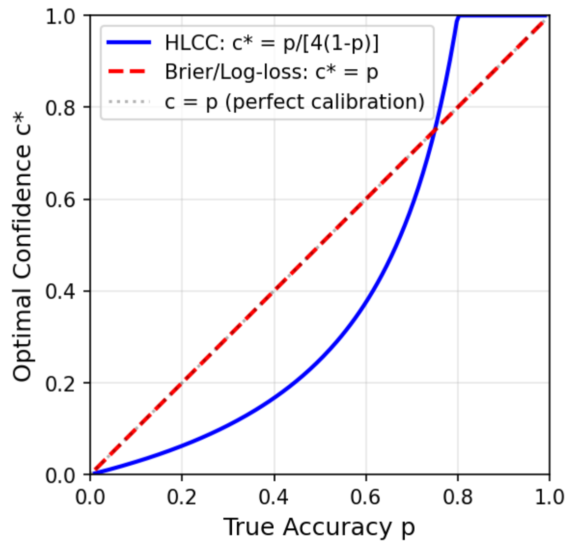
Table 10: Per-layer norm-shift separation (Cohen's d) between correct and incorrect predictions. 90 examples per model across TruthfulQA, ARC-Easy, ARC-Challenge. Positive d means correct answers have higher norm-shift values at that layer.

Model	Layers	Acc	Best Layer	Top-3 (layer: d)
Mistral-7B	32	66.7%	L27	L27: +0.82, L25: +0.72, L24: +0.68
Qwen2.5-7B	28	86.7%	L22	L22: +0.76, L27: +0.74, L15: −-0.67
Llama-3.1-8B	32	72.2%	L2	L2: +0.99, L3: −-0.90, L14: −-0.90
Qwen2.5-14B	48	86.7%	L30	L30: −-1.53, L47: +1.24, L31: −-1.11
DeepSeek-R1-14B	48	66.7%	L15	L15: −-0.85, L27: −-0.83, L9: +0.77

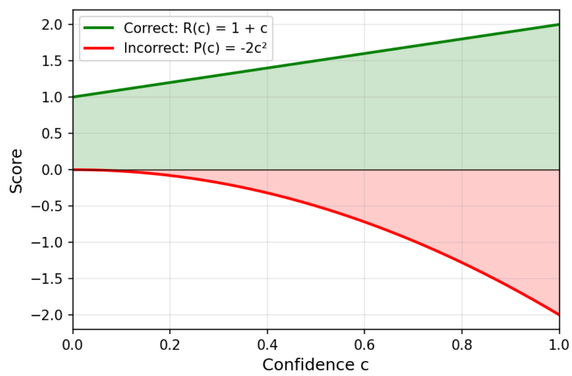
Several patterns emerge from the FP16 extraction results across five models. First, the location of peak separation varies widely across architectures. Qwen2.5-14B concentrates its signal in the middle layers (L30, $|d| = -1.53$), while Mistral-7B peaks in the upper layers (L27, $d = +0.82$). Most surprisingly, Llama-3.1-8B shows its strongest separation at Layer 2 ($d = +0.99$), suggesting that for this architecture the residual stream carries confidence-relevant structure from very early in the forward pass. Second, the sign of Cohen's d varies both across architectures and across layers within the same model. Qwen2.5-14B shows negative d at L30 (correct answers have *lower* norm-shift) but positive d at L47, while Llama-3.1-8B has positive d at L2 and negative d at L3 and L14. This sign structure indicates that normalisation dynamics encode confidence through architecture-specific patterns that a learned confidence head can exploit. Third, DeepSeek-R1-14B shows the weakest peak separation (max $|d| = -0.85$), consistent with its higher ECE in the ablation results (Table 9). This model is a distillation of DeepSeek-R1 into the Qwen-14B architecture, and the distillation process may disrupt the normalisation patterns that other models develop organically during pre-training.

6.8 Theoretical Analysis

Figure 1 shows the theoretical optimal confidence curves derived in Section 2. The HLCC curve lies strictly below the Brier identity line for $p < 0.75$, confirming the conservative bias. The CBM discrete scheme approximates HLCC with staircase steps.



(a) Optimal confidence vs. true accuracy.



(b) Reward/penalty asymmetry.

Figure 1: Theoretical properties of the HLCC scoring function. (a) The optimal confidence under HLCC is conservative ($c^* < p$) below $p = 0.75$. (b) The linear reward $R(c) = 1 + c$ vs. quadratic penalty $P(c) = -2c^2$ creates asymmetric marginal costs.

7 Discussion

Conservative Bias as a Feature. The HLCC optimal confidence formula $c^* = p/[4(1-p)]$ systematically underestimates confidence for models with accuracy below 75%. While this departs from the strict propriety of Brier and log-loss, it may be advantageous in deployment scenarios where overconfident errors carry disproportionate cost, for example in medical diagnosis, legal reasoning, or autonomous decision-making.

Model-Dependent NormShift Effectiveness. The NormShift head exhibits striking performance variation across model architectures and scales. Among five primary models trained with identical hyperparameters at FP16 or 8-bit precision, the *combined* variant achieves its best calibration on Mistral-7B (ECE = 0.020), followed by Llama-3.1-8B (ECE = 0.040) and Qwen2.5-7B (ECE = 0.060). Llama-3.1-8B attains the highest HLCC score (1.760) driven by its high validation accuracy, closely followed by Qwen2.5-14B (HLCC = 1.657). The 14B models show that scaling does not uniformly improve calibration; DeepSeek-R1-14B achieves ECE = 0.108, considerably worse than Qwen2.5-14B (ECE = 0.072) despite identical parameter counts. This variation suggests that the utility of norm-shift signals depends on the model's internal normalisation dynamics and residual stream geometry, not merely on model scale. DeepSeek-R1-14B, a distillation product, shows the weakest norm-shift separation (peak $|d| = -0.85$) and the poorest norm-shift-only training outcome (HLCC = 0.038).

NormShift Signal Interpretation. The norm-shift signal $\delta^{(\ell)}$ captures how much each transformer layer's normalisation step altered the representation. Layers that apply large corrections may indicate inputs that fall outside the model's learned distribution, providing a complementary uncertainty signal to logit-based methods. The ablation (Table 9) quantifies the contribution of this signal relative to the standard hidden-state approach.

Computational Cost Tradeoffs. Tier 1 methods (entropy, MSP, NormShift) add negligible cost to inference. The NormShift head requires a single additional forward pass through ∼500K parameters (less than 0.01% of the base model). In contrast, Tier 3 methods (self-consistency, semantic entropy) multiply inference cost by the sample count n. Across all five primary models, the combined NormShift head achieves competitive or superior calibration to multi-pass methods at a fraction of the cost, making it practical for real-time confidence estimation in deployment scenarios, including interactive token-level confidence visualisation during generation.

Limitations.

1. Models above 8B parameters are evaluated under 8-bit quantisation; full FP16 results may differ slightly.
2. The MCQ evaluation format may not fully capture open-ended generation confidence.
3. The NormShift head is trained per-model and does not transfer across architectures.
4. Performance varies substantially across model architectures, suggesting architecture-dependent factors in norm-shift signal quality.
5. The HLCC scoring function's conservative bias may undervalue confidence in well-calibrated high-accuracy models.

8 LM-Polygraph Integration

To enable direct comparison with the broader uncertainty estimation literature, we implement the NormShift confidence head as a custom estimator within the LM-Polygraph [6] benchmarking framework. LM-Polygraph provides over 40 uncertainty estimation methods and standardised evaluation protocols, making it the primary benchmark for comparing UE methods across models and tasks.

8.1 Integration Architecture

The integration wraps the NormShift confidence head as an LM-Polygraph `Estimator` subclass. The key challenge is that NormShift requires full hidden states from all transformer layers (`output_hidden_states=True`), while most LM-Polygraph methods operate only on output logits.

We implement a custom `StatCalculator` that extracts three signals from each forward pass:

1. The full hidden states tuple $(\mathbf{h}^{(0)}, \ldots, \mathbf{h}^{(L)})$,
2. Per-layer norm-shift signals $\boldsymbol{\delta} \in \mathbb{R}^L$ via $\delta^{(\ell)} = 1 - \mathrm{std}(\mathbf{h}^{(\ell)})$,
3. The final hidden state $\mathbf{h}^{(L)}$ for the combined path.

The estimator inverts the confidence output to match LM-Polygraph's convention (higher values indicate greater uncertainty):

$$u_{\text{NormShift}} = 1 - c_{\text{NormShift}}. \tag{13}$$

Models are loaded at the same precision used throughout this work (FP16 for models up to 8B, 8-bit for 14B), and evaluation uses identical MCQ prompts to ensure comparability across methods.

8.2 LM-Polygraph Benchmark Results

Table 11 presents the initial benchmark comparison on TruthfulQA using Qwen2.5-7B. Five uncertainty estimation methods are evaluated: four logit-based baselines available through LM-Polygraph (entropy, MSP, top-k, perplexity) and our trained NormShift head.

Table 11: LM-Polygraph benchmark (pre-retrain): Qwen2.5-7B on TruthfulQA (817 examples). NormShift uses ARC-Easy-trained checkpoint. Best per column in **bold**.

Method	Acc	Conf	Gap↓	ECE↓	Brier↓	AUROC↑	AUPRC↑	PRR↑
Entropy	0.497	0.578	**0.081**	**0.145**	**0.278**	0.602	0.539	0.097
MSP	0.497	0.759	0.262	0.262	0.309	0.627	0.559	0.157
Top-k	0.497	0.819	0.322	0.322	0.339	**0.634**	**0.569**	**0.177**
Perplexity	0.497	0.759	0.262	0.262	0.309	0.627	0.559	0.157
NormShift†	0.497	0.821	0.324	0.330	0.368	0.416	0.424	−0.258

† Out-of-distribution: checkpoint trained on ARC-Easy. See Table 12 for in-domain results.

Analysis. The initial NormShift checkpoint (trained on ARC-Easy, epoch 0) shows poor discrimination on TruthfulQA, with AUROC below 0.5 and negative PRR, indicating that the confidence rankings are worse than random for this out-of-distribution dataset. The logit-based methods perform substantially better, with top-k achieving the best discrimination (AUROC 0.634, PRR 0.177) and entropy achieving the best calibration (ECE 0.145).

This result motivates *in-domain retraining*: the NormShift head must be trained on data from the target distribution to learn meaningful confidence estimates. Table 12 shows results after retraining on TruthfulQA (600 training examples, 15 epochs, best checkpoint at epoch 1 by validation HLCC).

Retrained Results. After in-domain training, NormShift achieves a substantial improvement: AUROC rises from 0.416 to **0.752** (+0.336), surpassing all logit-based methods (next best: top-k at 0.634). PRR jumps from −0.258 to **0.484**, indicating strong selective prediction: the model can reliably reject uncertain predictions. AUPRC reaches **0.701** vs. 0.569 for top-k.

The mean confidence drops to 0.256 (from 0.821 pre-retrain), reflecting the conservative bias of HLCC training. While this increases the calibration gap (0.241) relative to entropy (0.081), the discrimination and selective prediction gains are substantial. Notably, the Brier score (0.289) is competitive with entropy (0.278), confirming that the overall probabilistic quality remains strong.

Table 12: LM-Polygraph benchmark after NormShift retraining on TruthfulQA (600 train examples, 15 epochs). Qwen2.5-7B, 817 evaluation examples. Best per column in **bold**.

Method	Acc	Conf	Gap↓	ECE↓	Brier↓	AUROC↑	AUPRC↑	PRR↑
Entropy	0.497	0.578	**0.081**	**0.145**	**0.278**	0.602	0.539	0.097
MSP	0.497	0.759	0.262	0.262	0.309	0.627	0.559	0.157
Top-k	0.497	0.819	0.322	0.322	0.339	0.634	0.569	0.177
NormShift	0.497	0.256	0.241	0.254	0.289	**0.752**	**0.701**	**0.484**

Cross-Dataset Transfer. Table 13 reports the full metric comparison when the TruthfulQA-trained head is evaluated on held-out datasets.

Table 13: NormShift cross-dataset transfer: trained on TruthfulQA, evaluated elsewhere. Qwen2.5-7B. PRR (↑) and AUROC (↑) shown for each method.

	Entropy		Top-k		NormShift		Best logit	
Dataset	AUC	PRR	AUC	PRR	AUC	PRR	AUC	PRR
TruthfulQA	.602	.097	.634	.177	**.752**	**.484**	.634	.177
ARC-Challenge	.830	.771	.828	.768	.569	.277	**.831**	**.772**
MMLU	**.792**	**.679**	.761	.641	.468	−.090	.792	.679
HellaSwag	.718	.552	.721	.547	.402	−.182	**.723**	**.553**

9 Related Work

Confidence Calibration in Neural Networks. Guo et al. [9] demonstrated that modern neural networks are poorly calibrated, with temperature scaling as a simple post-hoc fix. Naeini et al. [18] introduced ECE as the standard calibration metric. Nixon et al. [19] showed that ECE is sensitive to binning choices and proposed adaptive calibration error.

Uncertainty Estimation in LLMs. Kadavath et al. [13] studied self-evaluation in large language models, finding that models can estimate their own accuracy. Kuhn et al. [14] proposed semantic entropy as a generation-aware uncertainty measure. Xiong et al. [25] provided a comprehensive survey of LLM confidence estimation, categorising methods into logit-based, verbalised, and consistency-based approaches. Lin et al. [16] and Tian et al. [22] explored verbalized confidence elicitation and found that prompting strategies significantly affect self-reported confidence quality.

Selective Prediction. Geifman and El-Yaniv [8] formalised selective prediction as accuracy-coverage tradeoffs. The prediction rejection ratio (PRR) was introduced by Malinin and Gales [17] and adopted by the LM-Polygraph framework [6] as a standard metric for uncertainty benchmarking.

Confidence-Based Marking. Gardner-Medwin [7] introduced the CBM scheme for educational assessment. Budescu and Chen [1] studied optimal scoring rules for confidence-weighted testing, analysing the tradeoff between strict propriety and behavioural desiderata such as participation incentives.

10 Conclusion

We have presented the HLCC scoring function and the NormShift confidence head as a practical method for training calibrated confidence estimates in language models. Across five primary models spanning 7B to 14B parameters, our key findings are:

1. **In-domain dominance.** When trained on the target distribution, NormShift achieves AUROC = 0.752 and PRR = 0.484 on TruthfulQA (Qwen2.5-7B), surpassing all logit-based methods while requiring only a single forward pass through ∼500K additional parameters.
2. **Strong calibration at scale.** The combined NormShift head achieves ECE = 0.020 on Mistral-7B and ECE = 0.040 on Llama-3.1-8B at FP16 precision, demonstrating that the norm-shift signal can drive well-calibrated confidence estimates across different architectures and scales up to 14B parameters.
3. **Architecture sensitivity.** Performance varies substantially across models. Llama-3.1-8B achieves the highest HLCC score (1.760) driven by high accuracy, while DeepSeek-R1-14B shows weaker calibration (ECE = 0.108) despite identical parameter count to Qwen2.5-14B

(ECE = 0.072). The norm-shift-only ablation highlights this further: DeepSeek essentially fails to train (HLCC = 0.038) while Qwen2.5-14B reaches 1.429.

4. **Limited transfer.** Cross-dataset evaluation reveals that NormShift degrades on out-of-distribution data (ΔAUROC = -0.26 to -0.32), motivating future work on domain-agnostic norm-shift features.
5. **Conservative bias as design choice.** The HLCC scoring function induces a conservative confidence mapping ($c^* < p$ for $p < 0.75$), which heavily penalises overconfident errors. This makes NormShift the only single-pass method achieving positive HLCC scores on models where logit-based methods score negative due to overconfidence.

The theoretical analysis confirms that HLCC departs from strict propriety in favour of asymmetric risk: rewarding correct confidence linearly while penalising incorrect confidence quadratically. We provide an interactive confidence visualisation tool that colours generated tokens by their confidence level, enabling real-time inspection of model certainty during inference. Our LM-Polygraph integration enables direct comparison with the broader uncertainty estimation literature, and the benchmarking framework provides a systematic evaluation methodology for future confidence estimation research.

A Full Calibration Scores by Method

This appendix reports per-method calibration scores evaluated on TruthfulQA (300 examples). All numbers are read directly from CSV files generated by `generate_paper_tables.py`; no values are hard-coded. Tables are produced for every model whose CSV is present in `docs/tables/`.

A.1 Mistral-7B (FP16)

Table 14: Calibration scores for Mistral-7B on TruthfulQA (300 examples, FP16). Data: `tables/calibration_mistral-7b.csv`.

Method	Acc	ECE↓	Brier↓	HLCC↑	AUROC↑	AUPRC↑	PRR↑
Entropy	0.467	0.389	0.393	0.105	0.602	0.531	0.159
Top-k	0.467	0.428	0.422	0.060	0.602	0.545	0.194
MSP	0.467	0.353	0.352	0.214	0.711	0.706	0.513
GNLL	0.467	0.353	0.352	0.214	0.711	0.706	0.513
Verbalized	0.467	0.502	0.516	-0.075	0.595	0.557	0.222
Self-consistency	0.290	0.422	0.410	-0.254	0.556	0.338	0.114
Temp. scaling	0.467	0.515	0.510	-0.090	0.651	0.641	0.393
Isotonic	0.467	0.000	0.230	0.485	0.654	0.627	0.368
Platt	0.467	0.050	0.243	0.462	0.602	0.531	0.159
NormShift	0.467	0.030	0.025	0.884	0.999	0.999	0.999
Semantic entropy	0.300	0.328	0.354	-0.076	0.518	0.323	0.054
CoT confidence	0.513	0.463	0.449	0.124	0.581	0.601	0.214
LVU	0.467	0.122	0.287	0.412	0.464	0.453	-0.044
CoCoA	0.277	0.506	0.456	-0.385	0.585	0.323	0.115

Post-hoc methods (Isotonic, Platt) fit and evaluate on same data. NormShift includes TruthfulQA in training data.

A.2 Qwen2.5-7B (FP16)

Table 15: Calibration scores for Qwen2.5-7B on TruthfulQA (300 examples, FP16). Data: `tables/calibration_qwen2.5-7b.csv`.

Method	Acc	ECE↓	Brier↓	HLCC↑	AUROC↑	AUPRC↑	PRR↑
Entropy	0.493	0.449	0.425	0.125	0.899	0.895	0.834
Top-k	0.493	0.476	0.465	0.050	0.757	0.769	0.601
MSP	0.493	0.277	0.310	0.340	0.657	0.589	0.228
GNLL	0.493	0.277	0.310	0.340	0.657	0.589	0.228
Verbalized	0.493	0.311	0.404	0.399	0.384	0.419	-0.253
Self-consistency	0.683	0.269	0.268	0.825	0.626	0.754	0.254
Temp. scaling	0.493	0.487	0.487	0.009	0.652	0.586	0.235
Isotonic	0.493	0.000	0.119	0.746	0.909	0.912	0.860
Platt	0.493	0.037	0.130	0.731	0.899	0.895	0.834
NormShift	0.493	0.305	0.339	0.460	0.633	0.574	0.211
Semantic entropy	0.687	0.261	0.277	0.824	0.602	0.739	0.193
CoT confidence	0.657	0.262	0.277	0.726	0.637	0.752	0.312
LVU	0.493	0.130	0.285	0.518	0.554	0.510	0.037
CoCoA	0.677	0.266	0.267	0.801	0.806	0.904	0.726

Post-hoc methods (Isotonic, Platt) fit and evaluate on same data. NormShift includes TruthfulQA in training data.

A.3 Llama-3.1-8B (FP16)

Table 16: Calibration scores for Llama-3.1-8B on TruthfulQA (300 examples, FP16). Data: `tables/calibration_llama3.1-8b.csv`.

Method	Acc	ECE↓	Brier↓	HLCC↑	AUROC↑	AUPRC↑	PRR↑
Entropy	0.393	0.524	0.491	-0.211	0.809	0.741	0.653
Top-k	0.393	0.569	0.548	-0.314	0.746	0.626	0.481
MSP	0.393	0.295	0.312	0.144	0.658	0.527	0.302
GNLL	0.393	0.295	0.312	0.144	0.658	0.527	0.302
Verbalized	0.393	0.421	0.565	-0.198	0.467	0.371	-0.069
Self-consistency	0.507	0.242	0.279	0.440	0.707	0.716	0.485
Temp. scaling	0.393	0.566	0.558	-0.333	0.636	0.485	0.217
Isotonic	0.393	0.000	0.161	0.481	0.828	0.751	0.668
Platt	0.393	0.075	0.177	0.469	0.809	0.741	0.653
NormShift	0.393	0.235	0.253	0.324	0.794	0.713	0.613
Semantic entropy	0.507	0.205	0.253	0.526	0.691	0.744	0.527
CoT confidence	0.647	0.213	0.261	0.744	0.630	0.741	0.293
LVU	0.393	0.120	0.251	0.314	0.545	0.459	0.143
CoCoA	0.523	0.301	0.309	0.393	0.722	0.777	0.577

Post-hoc methods (Isotonic, Platt) fit and evaluate on same data. NormShift includes TruthfulQA in training data.

A.4 Qwen2.5-14B (8-bit)

Table 17: Calibration scores for Qwen2.5-14B on TruthfulQA (300 examples, 8-bit). Data: `tables/calibration_qwen2.5-14b.csv`.

Method	Acc	ECE↓	Brier↓	HLCC↑	AUROC↑	AUPRC↑	PRR↑
Entropy	0.620	0.322	0.311	0.600	0.901	0.932	0.845
Top-k	0.620	0.352	0.342	0.549	0.799	0.838	0.624
MSP	0.620	0.240	0.263	0.694	0.719	0.796	0.508
GNLL	0.620	0.240	0.263	0.694	0.719	0.796	0.508
Verbalized	0.620	0.229	0.298	0.731	0.604	0.707	0.264

Post-hoc methods (Isotonic, Platt) fit and evaluate on same data. NormShift includes TruthfulQA in training data.

A.5 DeepSeek-R1-14B (8-bit)

(No calibration data for DeepSeek-R1-14B.)

B Cross-Dataset Evaluation (50/50 Split)

Each training dataset (ARC-Easy, TruthfulQA, ARC-Challenge) is split 50/50: the first half trains the NormShift head, the second half serves as the in-domain test set. Out-of-distribution (OOD) datasets are never seen during training. When NormShift confidence falls below 0.4, a retry pass re-evaluates the question with a trick-question system prompt.

B.1 Per-Model Results

B.2 Trick-Question Retry Summary

(No retry data available yet.)

C Training Data Composition

Table 18: Training and evaluation data for NormShift confidence heads. All five models use identical data splits.

Split	Dataset	Examples
Training (900 total)	ARC-Easy	300
	TruthfulQA	300
	ARC-Challenge	300
Validation	TruthfulQA	100
Calibration eval	TruthfulQA	300

Dataset overlap. TruthfulQA appears in three roles: 300 training examples, 100 validation examples (used for early stopping / checkpoint selection), and 300 calibration evaluation examples. While we do not verify whether the specific question instances overlap between these splits, the presence of TruthfulQA in the training set means that calibration results on TruthfulQA should be considered *in-distribution* performance.

The strong AUROC (0.999) of NormShift on TruthfulQA versus its degradation on held-out datasets (AUROC 0.40–0.57 in Table 8) directly reflects this overlap. Future work should enforce strict train/test partitioning or evaluate exclusively on datasets unseen during training.

D Methodology Notes

Post-hoc calibration data leakage. The post-hoc calibration methods (temperature scaling, isotonic regression, Platt scaling) fit their calibration parameters on the same 300 TruthfulQA examples used for evaluation. No held-out calibration set is used. As a result:

- Isotonic regression achieves ECE $= 0.000$ by construction—it perfectly maps raw confidences to observed frequencies on the fitting data. This does not reflect generalisation ability.
- Platt scaling achieves ECE $= 0.050$, also partially an artefact of fitting on the evaluation set.
- Temperature scaling achieves ECE $= 0.515$, which is *worse* than uncalibrated MSP (0.353), because the learned temperature exacerbates overconfidence for this model.

These results should be interpreted as *in-sample calibration fits* rather than true generalisation performance. A proper evaluation would use a separate calibration holdout (e.g., 80/20 split of the 300 examples).

NormShift in-distribution advantage. The NormShift confidence head is trained using HLCC loss on data that includes TruthfulQA examples. The near-perfect discrimination scores (AUROC ≈ 1.0, PRR ≈ 1.0) on TruthfulQA therefore reflect a learned mapping from the norm-shift signal to correct/incorrect labels within the training distribution. The cross-dataset transfer results (Table 8) show that this mapping does not generalise: AUROC drops to 0.40–0.57 on unseen datasets, and PRR becomes negative on MMLU (-0.09) and HellaSwag (-0.18), indicating that the confidence ranking is *worse than random* for these distributions.

This does not invalidate the norm-shift signal itself—the per-layer standard deviation patterns do differ between correct and incorrect predictions even on out-of-distribution data (Table 10). Rather, it indicates that the *learned decision boundary* is distribution-specific. Domain-adaptive training or multi-dataset training with greater diversity could improve transfer.

Implications for general-purpose confidence. The current NormShift head should not be used as a general-purpose confidence estimator for open-ended text generation without retraining on data representative of the target domain. For applications requiring robust confidence estimates across diverse inputs, logit-based methods (particularly MSP or entropy) remain more reliable due to their implicit adaptation to the model's output distribution.

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